UNIT-2

Chapter-2. RELATIONS

1. RELATIONS ON SETS:

To express a relationship between elements of two sets to use ordered pair (Cartesian Product) made up of two related elements. Let A and B be two sets as follows:

A = { Calcutta, Patna, Lucknow, Chennai }

B = { West Bengal, Bihar, Uttar Pradesh, Tamil Nadu }

Relation between set A and set B is given by Ordered pair.

$A \times B = R$ = {(Calcutta, West Bengal), (Patna, Bihar), (Lucknow, Uttar Pradesh), (Chennai, Tamil Nadu)}

$A \times B = R = \{(x, y) : x \in A, y \in B, xRy\}$

2. Definition (Relation):

Any set of ordered pair defined a relation. Let A and B are two sets then A Relation from A to B is a subset of the cartesian product $A \times B$. Suppose R is the relation from A to B. Then R is a set of ordered pair (a, b) where $a \in A$, $b \in B$, every such ordered pair is written as aRb and read a related to b. by R. If $(a, b) \notin R$, then a is not related to b.

and written as $a R b$ is called a binary relation from A to B.

Relation is called by Cross Product of AX consists of all ordered pairs whose First element is in A and Second element is in

$$R = A \times B = \{(x, y) : x \in A \text{ and } y \in B\}$$

Ex1. If $A = \{1, 2, 5\}$, $B = \{2, 4\}$ then

$$R = A \times B = \{(1, 2), (1, 4), (2, 2), (2, 4), (5, 2), (5,$$

Ex2 Find relation between A and B if x<

$$R = \{(1, 2), (1, 4), (2, 4).\}$$

Ex3. If R is relation from set A to itself

$$R = A \times A = A^2 = \{(x, x) : x \in A \text{ and } x \in A\}$$

Relation of R on. A.

3. DOMAIN AND RANGE

Let A and B are two non-empty sets. R be a binary relation from A to B.

Then $\{a \in A$ and $b \in B$ then $R = \{(a, b)\}$

① Domain of $R = \{a, (a, b) \in R\}$

② Range of $R = \{b, (a, b) \in R\}$

① Domain is the first elements of R

② Range is the Second elements of R.

Solved Examples:

Example 1. Let $A=\{2,3,5\}$, $B=\{2,4,6,10\}$ A Relation of R from A to B. is given as follows $2R2, 2R4, 2R6, 3R6, 5R10$

Write R is a set of ordered pairs

Solution: $\{(2,2),(2,4),(2,6),(3,6),(5,10)\}$

Example 2 Let $A=\{2,3,4\}$, $B=\{3,4,5\}$

List the elements of each relation R defined below and the Domain and Range.

(i) $a \in A$ is related to B. $b \in B$ that is aRb if $a<b$

(ii) $a \in A$ is related to B $b \in B$ that is aRb if a and b both odd numbers

Solution $R=\{(2,3),(2,4),(2,5),(3,4),(3,5),(4,5)\}$

$Dom(R)=\{2,3,4\}$ and Ran of $R=\{3,4,5\}$

(ii) $R=\{(3,3),(3,5)\}$

$Dom(R)=\{3\}$

$Range(R)=\{3,5\}$

Example 3 let $S=\{x,y\}$ and S^2 is Cartesian Product between $S \times S$.

(i) Find elements of $S^2 = S \times S = \{x,y\} \times \{x,y\}$

$= \{(x,x),(x,y),(y,x),(y,y)\}$

(ii)

TOTAL NUMBER of DISTINCT RELATION FROM a SET A to a SET B.

Let the number of elements of A and B are m and n repectively. Then the number of elements of $A \times B$ is mn. The number of elements of the sets $A \times B$ is $2^{m.n}$. Now every subset of $A \times B$ is a relation from A to B.

for example if $A=\{1,2,3\}$ and $B=\{2,4,5\}$ then the number of relations between A to B

$= 2^{m.n} = 2^{3 \times 3} = 2^9 = 512$

SET OF OPERATIONS ON RELATIONS

① If R and S are two relations then

$R \cap S \iff x(R \cap S)y \iff xRy \wedge xSy$

UNIT-2 (Relations and functions)

1. **Relation On Sets**: The most direct way to express a relationship between elements of two sets is to used ordered pair maid up of two related elements.

let A and B two sets as follows

$A = \{$ Calcutta, Patna, Lucknow$\}$ and

$B = \{$ West Bengal, Bihar, Uttarpradesh$\}$

$R = \{$(Calcutta, West Bengal), (Patna, Bihar), (Lucknow, Uttarpradesh)$\}$

$R = \{(x, y) \Rightarrow x \in A, y \in B, R(x,y)\} \Rightarrow xRy$

this is represent by $A \times B$.

$R = A \times B = \{(x, y)\}$ where $(x, y) \in R$

Definition: Any set of ordered pairs defined a binary relation or relation.

let A and B be two sets then A relation from A to B is a subset of Cartesian Product $A \times B$. Suppose R is a relation from A to B. then R is a set of ordered pairs (a, b) where $a \in A$ and $b \in B$ Every ordered pair is written as $aRb \Rightarrow$ a related to b

If $(a,b) \notin R$, then a is not related with B. $\Rightarrow a \not R b$. then $a \not R b$ is called a binary relation from A to B.

Cartesian Product: $A \times B$ has all ordered pairs whose first element in A and second element in B.

$R = A \times B = \{(x,y) : x \in A \text{ and } x \in B\}$

If $A = \{1,2,5\}$, $B = \{2,4\}$

then $A \times B = \{(1,2),(1,4),(2,2),(2,4),(5,2),(5,4)\}$

If we take a relation $x < y$

$R = \{(1,2),(1,4),(2,4)\}$, $x < y$,

This is called relation of R from A to B

If both sets are same then this is called Relation of R on A

$A^2 = A \times A = \{1,2,5\} \times \{1,2,5\}$

$= \{(1,1),(1,2),(1,5),(2,1),(2,2),(2,5)$
$(5,1),(5,2),(5,5)\}$

Domain and Range:

Let A and B two non-empty sets. then R be the binary relation from A to B.

The set $\{a \in A : (a,b) \in R \text{ and } b \in B\}$ is called the domain of R and denoted by dom(R)

The Set $= \{ b \in B : (a, b) R \text{ and } a \in A \}$ is called range of R and denoted by Ran(R)

Example-1. let $A = \{2, 3, 5\}$, $B = \{2, 4, 6, 10\}$ A relation R from A to B is given as follows $2R2$, $2R4$, $2R10$, $5R10$, Write R is a set of ordered pair.

Solution $R = \{(2, 2), (2, 4), (2, 6), (2, 10), (3, 6), (5, 10)\}$

$R = \{(2, 2), (2, 4), (2, 6), (2, 10), (3, 2), (3, 4), (3, 6), (3, 10), (5, 2), (5, 4), (5, 6), (5, 10)\}$

Example 2 :- let $A = \{2, 3, 4\}$, $B = \{3, 4, 5\}$

(i) $a \in A$ is related to $b \in B$, that aRb if $a < b$

$R = \{(2, 3), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5)\}$

Dom $R = \{2, 3, 4\}$, Range $(R) = \{3, 4, 5\}$

(ii) $a \in A$ is related to $b \in B$. that aRb if

are odd numbers.

Example 4. $A = \{1, 2, 3, 4, 5, 6\}$ and R be the relation on A defined by $(a, b) \in R$ if and only if a is a multiple of b. Write down R as a set of ordered pair.

Solution $R = \{(1,1), (2,1), (2,2), (3,1), (3,3), (4,1), (4,2), (4,4), (6,1), (6,2), (6,3), (6,6)\}$.

Ex 5. Let $A = \{1, 2, 3, 4\}$ and R be the relation on A defined by $(a, b) \in R$ if and only if a divides b. Write down R as set of ordered pair.

Solution $R = \{(1,1), (1,2), (1,3), (2,2), (1,4), (2,4), (3,3), (4,4)\}$.

Example 6. (i) Determine $A \times B$ given $A = (a, b)$ and $B = \{1, 2, 3\}$,

Ans $R = \{(a,1), (a,2), (a,3), (b,1), (b,2), (b,3)\}$

(ii) Let $A = \{0, 1, 2\}$ and $B = \{a, b\}$,

Find $A \times B$ and $B \times A$.

Solution $A \times B = \{(0,a), (0,b), (1,a), (1,b), (2,a), (2,b)\}$.

$B \times A = \{(a,0), (a,1), (a,2), (b,0), (b,1), (b,2)\}$

$A \times B \neq B \times A$.

Example 7: for which Set A, and B the relation $A \times B = B \times A$ hold.

Solution: In generally. $A \times B \neq B \times A$

The relation $A \times B = B \times A$ holds only. When A or B is a null set. Suppose $A = \phi$

$$\phi \times B = B \times \phi = \phi.$$

SET OPERATIONS ON RELATIONS.

If R and S be any two Relations then

1. Intersection of R and S $\Rightarrow$ $R \cap S$ is also a relation. and

$x(R \cap S)y \Leftrightarrow xRy \wedge xSy$

2. Union of R and S. (i.e.,) $R \cup S$. or

is a relation such that

$x(R \cup S)y \Leftrightarrow xRy \vee xSy$

3. Difference of R and S.

$x(R - S)y \Leftrightarrow xRy \wedge xSy$

4 Complement of relation R

$x(R')y = x \not R y$. where R' is Complement of R.

Example 9. If $A=\{x,y,3\}$, and $B=\{X,Y,Z\}$, $C=\{x,y\}$, and $D=\{Y,Z\}$. R is a relation from A to B. defined by R

$$R=(A\times B)=\{(x,X),(x,Y),(x,Z),(y,X),(y,Y),(y,Z),(3,X),(3,Y),(3,Z)\}$$

$$S=(C\times D)=\{(x,Y),(x,Z),(y,Y),(y,Z)\}$$

$R=\{(x,X),(x,Y),(y,Z)\}$ — given

$S=\{(x,Y),(y,Z)\}$ — given

TYPES OF RELATION IN A SET.

1. **Inverse Relation**: Let R be any relation between A to B then Inverse relation b/w R^{-1}: B to A.

$R=\{(a,b), a\in A \text{ and } b\in B\}$

$R^{-1}=\{(b,a)\ b\in A \text{ and } a\in B\}$

$xRy = yR^{-1}x$

2. **Identity Relation**:-

A relation R in a set A is said to be identity relation. denoted by I_A, if $I_A=\{(x,x): x\in A$

A relation from A to itself or $R\longrightarrow A\longrightarrow A$ is called Identity relation.

Ex $A=\{1,2,3\}$ then $I_A=\{(1,1),(2,2),(3,3)\}$. is an identity relation in A.

Ex10 let $A=\{2,3,5\}$ and $B=\{6,8,10\}$,
A relation R from A to B.
for all $(x,y)\in(A\times B)$, $(x,y)\in R \Leftrightarrow \frac{x}{y}$ (x divides y)
Write each R and R' as a set of ordered pairs.

Solution $2\in A$ divides $6\in B$ then $2R6$, $2R8$, $2R10$
$3R6$, $5R10$.

$R=\{(2,8),(2,10),(3,6),(5,10)\}$

$R^{-1}=\{(8,2),(10,2),(6,3),(10,5)\}$

Dom $R=\{2,3,5\}$
Rang of $R=$ Dom $(R^{-1})=\{8,10,6\}$.

$\Rightarrow (R^{-1})^{-1}=R$.

n-Ary Relation

Let $(A_1,A_2,A_3,A_4,\cdots,A_n)$ be a finite collection of sets. A subset R of $A_1\times A_2\times A_3\times\cdots\times A_n$ is called n-ary relation on $A_1,A_2,A_3,\cdots,A_n$.

(i) If $R=\phi$ then R is called Empty relation

(ii) If $R=A_1\times A_2\times A_3\times\cdots\times A_n$ then R is called Universal relation.

(iii) If $A_i=A$, $i=1,2,3,\cdots n$, then R is called n-ary relation of A.

(iv) for $n=1,2$, or 3 then R is called Unary, binary or Ternary rela

Ex1. $A = \{2, 5, 7\}$, $R \to A \times A$

$R = \{2, 5, 7\} \times \{2, 5, 7\} \Rightarrow a \neq b$

$= \{(2,5), (2,7), (5,2), (5,7), (7,2), (7,5)\}$.

Properties of Relations in A Set.

A relation R on a set A satisfy Cartesian Properties.

1. **Reflexive Relation:**

A relation R on set A is reflexive if aRa for $\forall a \in A$, if (a,a) it means each element of A related to itself.

2. **Symmetric relation:-**

$A = \{1,2,3\}$, $R_1 = \{(1,1), (1,2), (1,3), (2,1), (2,2), (3,1)\}$

3. **Transitive relation:-** $(a,b) \in R$ and $(b,c) \in R$ $(a,c) \in R$. aRb and bRc $\Rightarrow$ aRc $\in R$.

PARTIAL ORDER RELATION (PARTIAL ORDERING)

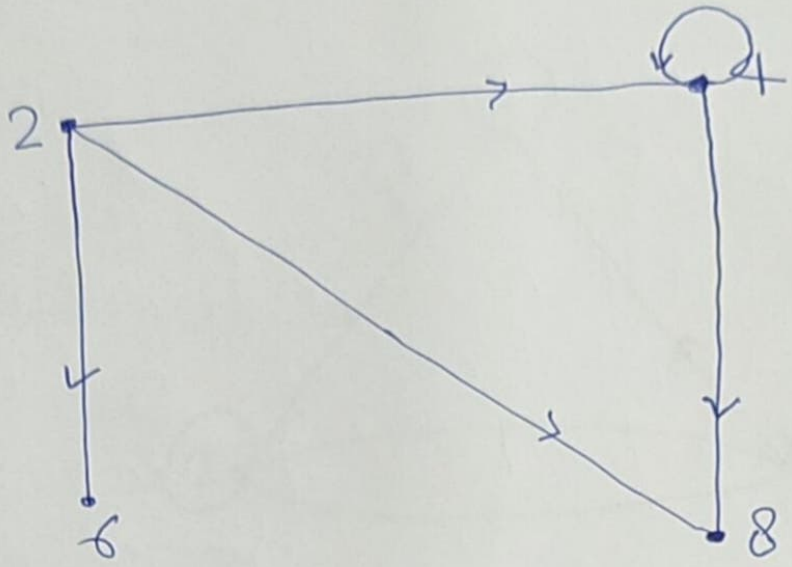
A relation R on a set A is called a partial order relation if and only if R is reflexive, Antisymmetric and transitive.

1. Reflexive $aRa, a \in A$
2. Antisymmetric $(a,b) \in R, (b,a) \notin R$
3. Transitive: $(a,b) \in R, (b,c) \in R$ then $(a,c) \in R$

GRAPH OF A RELATION:

Let A and B are two finite set and R is a relation from A to B. For Graphically representation of a relation on a set. each element of a set represented by a Point. These Points are called nodes or Vertices. An arc is drawn from each Point to its related Point. If the pair $x \in A$, $y \in B$ is in the relation. the corresponding nodes are connected by arc called edges or arcs. The arc start from First element of the pair and go to the second element of the pair. The direction indicated by an arrow. If $(x_i, x_j) \in R$ then nodes x_i and x_j by means of an arc and Put an arrow in the direction from x_i to x_j. A Point $(a,a) \in R$ is represented an arc from node a to itself. Such an edge is called a loop.

Ex1. let $A = \{2,4,6\}$, $B = \{4,6,8\}$ and R be the relation from A to B given by $x R y$ then $R = \{(2,4), (2,6), (2,8), (4,4), (4,6), (4,8), (6,6)\}$



(i) A relation R is reflexive because $(a,a) \in R$. So R is reflexive; $(4,4) \in R$

(ii) A relation R is said to symmetric if $(a,b) \in R$ then $(b,a) \in R$.

But Here we see that $(2,4) \in R$ and $(4,2) \notin R$ $(2,8) \in R$ but $(8,2) \notin R$, $(2,6) \in R$ but $(6,2) \notin R$

So relation R is not symmetric.

(iii) $(2,4) \in R$, $(4,8) \in R$ then here $(2,8) \in R$

So R is transitive.

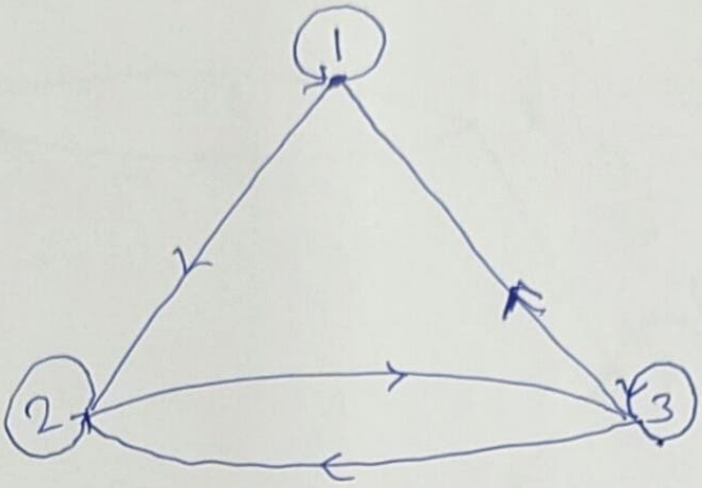
Finally the given relation is an equivelance relation.

Ex 2. Draw the directed graph that represented the relation

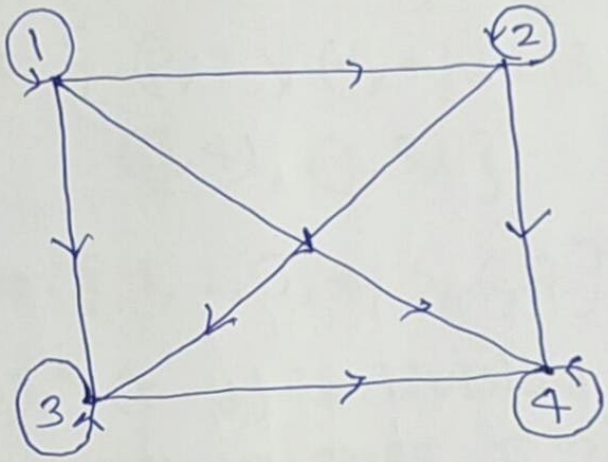
$R = \{(1,1), (2,2), (1,2), (2,3), (3,2), (3,1), (3,3)\}$

~~Where~~ were set $X = \{1,2,3\}$

Solution:



Example 2.



$R = \{(1,1),(2,2),(3,3),(4,4),(1,2),(2,4),(1,4),(2,3),(3,4)\}$

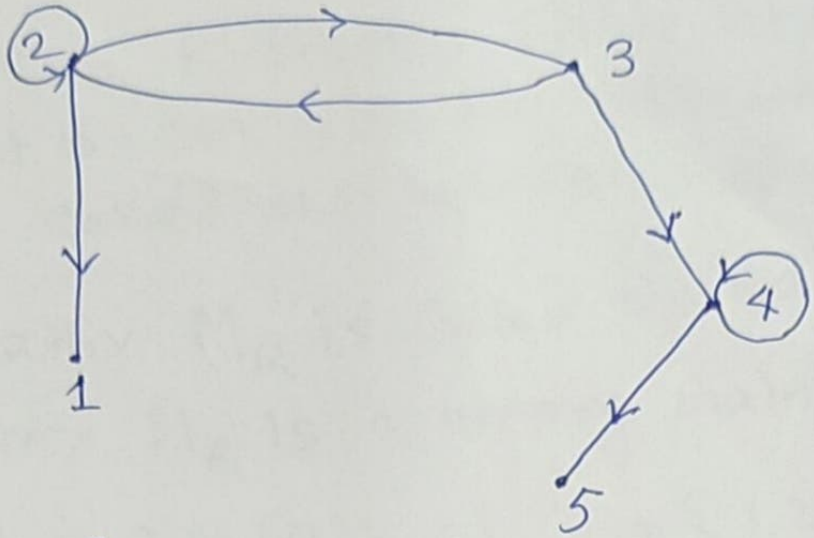
(1) R is reflexive because $(1,1) \in R$, $(2,3) \in R$, $(3,3) \in R$, $(4,4) \in R$.

(2) R is not symmetric because $(2,3) \in R$ but $(3,2) \notin R$, $(2,4) \in R$ but $(4,2) \notin R$. $(3,4) \in R$ but $(4,3) \notin R$.

(3) R is transitive because $(2,3) \in R$ and $(3,4) \in R$ then, $(2,4) \in R$.

So R is an equivalence relation.

Ex4.

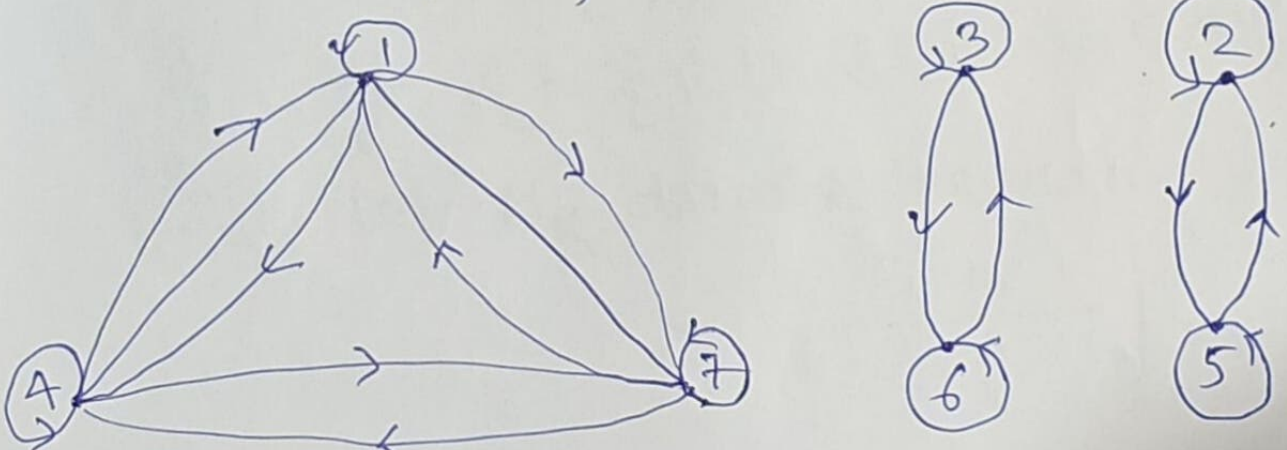


Solution $R = \{(2,2), (4,4), (2,3), (3,2), (2,1), (3,4), (4,5)\}$

Ex5 Let $X = \{1,2,3,4,5,6,7\}$ and $R = \{(x,y) \mid (x-y) \text{ is divisible by } 3\}$ Show that R is an equivalence relation. Draw the graph of R.

Solution $R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6), (7,7), (1,4), (4,1), (6,3), (3,6), (7,1), (1,7), (7,4), (4,7), (5,2), (2,5)\}$

Now Graph of R is



The Matrix of a relation:

A relation R from a finite set A to a finite set B can also be represented by a matrix called the relation matrix of R.

The matrix M_R is called the matrix of R.

The matrix M_R is a binary matrix.

Example 1. $A = \{a, b, c\}$, $B = \{1, 2, 3\}$ and $R = \{(a, 1), (a, 3), (b, 2), (c, 1)\}$ then the table is.

	1	2	3
a	1	0	1
b	0	1	0
c	1	0	0

Hence for given relation R, the relation matrix is $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

Ex2: Let R be the relation from the set A $A = \{1, 3, 4\}$, on itself and defined by $R = \{(1, 1), (1, 3), (3, 3), (4, 4)\}$

Sol. Let M_R denote the matrix of R.

	1	3	4
1	1	1	0
3	0	1	0
4	0	0	1

$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Ex3. let $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3, b_4\}$ Which ordered pairs are in the relation R represented by the matrix

$$M_R = \begin{array}{c} \\ a_1 \\ a_2 \\ a_3 \end{array} \begin{array}{c} \begin{array}{cccc} b_1 & b_2 & b_3 & b_4 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

Solution: Now R contains of those ordered pairs

$R = \{(a_1, b_2), (a_2, b_1), (a_2, b_3), (a_2, b_4), (a_3, b_1), (a_3, b_3)\}$.

Ex4: let $X = \{1, 2, 3, 4\}$ be a set X and R is a relation on the set X such that

$R = \{(1,1), (1,4), (4,1), (4,4), (2,2), (2,3), (3,2), (3,3)\}$

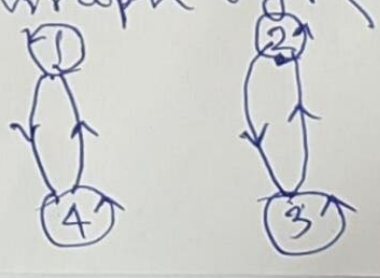
Find a matrix of relation and Draw the graph.

Solution:-

	1	2	3	4
1	1	0	0	1
2	0	1	1	0
3	0	1	1	0
4	1	0	0	1

$$M_R = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Graph of R



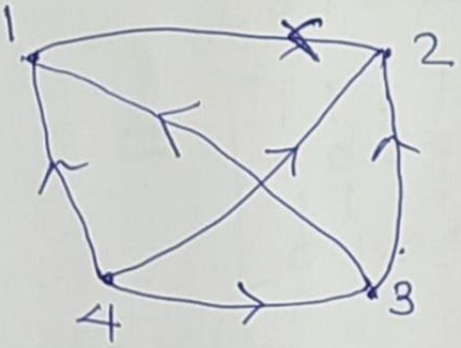
Example 4. Let $X = \{1, 2, 3, 4\}$, and Relation $R = \{(x, y) \mid x > y\}$ Draw the graph of R and also give its matrix.

Solution: From the definition of the given R, We note that $R = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$

By examining the elements of R, we find that the matrix of R is

$$M_R = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

The diagram of R is as shown.



Example 5. Let $A = \{1,2,3,4\}$ and R be the relation on A whose matrix is

~~$R = \{(1,1),(1,2),(1,3)$~~

$$M_R = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Show that R is transitive

Solution: $M_R^2 = M_R \cdot M_R$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

So. $M_R^2 + M_R = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = M_R.$$

① $M_{R \cup S} = M_R \vee M_S$

② $M_{R \cap S} = M_R \wedge M_S$

③ $M_{R^{-1}} = M_R^T$.

Where M_R^T is the transpose of M_R.

Solution: Here Relation matrix is given

$$M_R = \begin{array}{c} \\ a_1 \\ a_2 \\ a_3 \end{array} \begin{array}{c} \begin{array}{cccc} b_1 & b_2 & b_3 & b_4 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

Now Find relation of a matrix

$$R = \{(a_1, b_2), (a_2, b_1), (a_2, b_3), (a_2, b_4), (a_3, b_1), (a_3, b_3)\}.$$

Ex4. If $X = \{1, 2, 3, 4\}$ be a set and R is a relation on the set X. Such that

$$R = \{(1,1), (1,4), (4,1), (4,4), (2,2), (3,2), (2,3), (3,3)\}$$

Solution: Matrix on a relation.

$$M_R = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 1 & 0 & 0 & 1 \\ 2 & 0 & 1 & 1 & 0 \\ 3 & 0 & 1 & 1 & 0 \\ 4 & 1 & 0 & 0 & 1 \end{array} \quad (X \times X)$$

$$M_R = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Relation Matrix [The matrix of a relation]

Ex1. If $A=\{a,b,c\}$, $B=\{1,2,3\}$ and $R=\{(a,1),(a,3),(b,2),(c,1)\}$ then find relation matrix.

Solution

M_R	1	2	3
a	1	0	1
b	0	1	0
c	1	0	0

$$\Rightarrow M_R=\begin{bmatrix}1&0&1\\0&1&0\\1&0&0\end{bmatrix}$$

Ex2. Let R be the relation from the set A $A=\{1,3,4\}$, on itself and defined by $R=\{(1,1),(1,3),(3,3),(4,4)\}$ then Find matrix of R.

Solution:

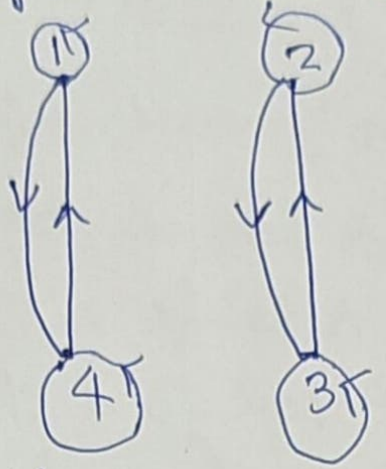
M_R = A	1	3	4
1	1	1	0
3	0	1	0
4	0	0	1

$$\Rightarrow M_R=\begin{bmatrix}1&1&0\\0&1&0\\0&0&1\end{bmatrix}$$

Ex3 If $A=\{a_1,a_2,a_3\}$ and $B=\{b_1,b_2,b_3,b_4\}$ and

$$M_R=\begin{bmatrix}0&1&0&0\\1&0&1&1\\1&0&1&0\end{bmatrix}$$

Graph on a relation.



① Relation is reflexive because
$(1,1) \in R$, $(2,2) \in R$, $(3,3) \in R$, $(4,4) \in R$

② Symmetric relation. because
$(1,4) \in R$, $(4,1) \in R$, $(2,3) \in R$, $(3,2) \in R$

③ Transitive relation:
$(2,3) \in R$, and $(3,2) \in R \Rightarrow (2,2) \in R$
$(1,4) \in R$ and $(4,1) \in R \Rightarrow (1,1) \in R$

Ex5. If $X = \{1,2,3,4\}$, and $R = \{(x,y) \mid x < y\}$

Solution
$R = X \times X = \{1,2,3,4\} \times \{1,2,3,4\}$
$R = \{(1,2), (1,3), (1,4), (2,3), (2,4), (3,4), \}$

Matrix of a relation.

	1	2	3	4
1	0	1	1	1
2	0	0	1	1
3	0	0	1	1
4	0	0	0	1

1. DEFINITION: FUNCTION (MAPPING)

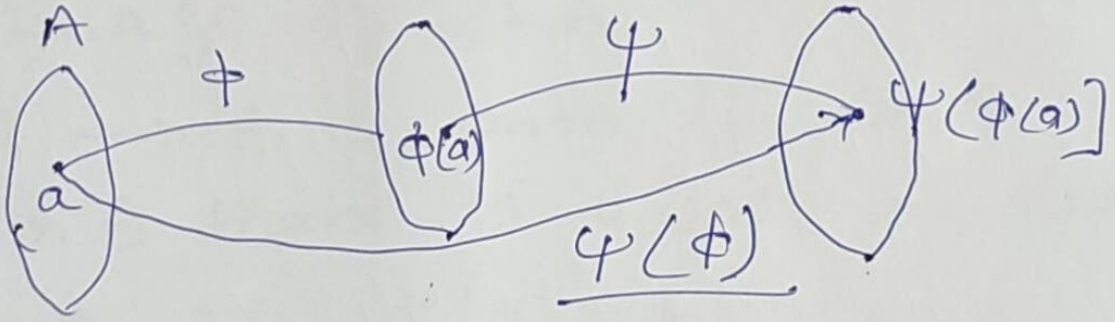
A function (or mapping) f from a set A to a set B is a rule that each element a of set A mapped exactly one element b of set B. then the set A is called domain of f and B is called Co-domain or Range of f.

$f: A \longrightarrow B$ means f is a mapping from A to B like $f(x) = y$, $x = a$, $y = b$

$$\boxed{f(a) = b}$$

2. Composition of the function

Let $f: A \longrightarrow B$. and $\phi: B \longrightarrow C$ the composition $f(\phi) \longrightarrow A \longrightarrow C$



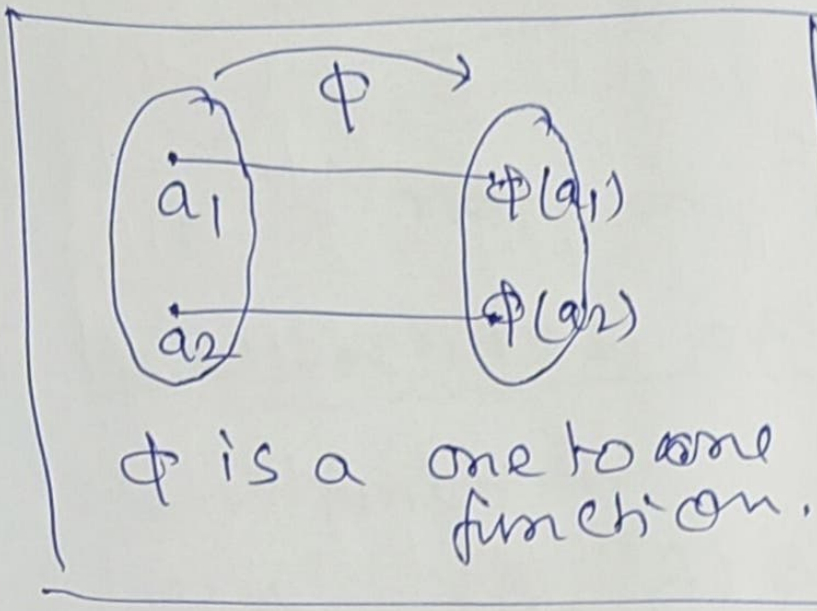
3. DEFINITION: ONE-to-ONE FUNCTION.

A function ϕ from a set A to a set B is called one to one

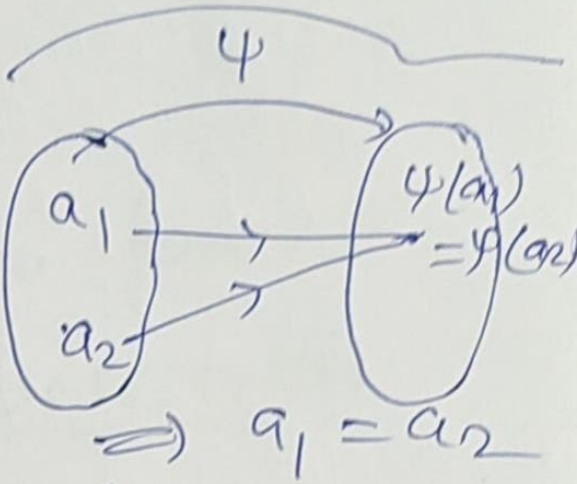
$$\phi(a_1) \neq \phi(a_2) \Longrightarrow a_1 \neq a_2$$

A function ψ from a set A to a set B is not one to one if $\psi(a) = \psi(b)$

$\Rightarrow \boxed{a_1 = b_1}$.



ϕ is a one to one function.



$\Rightarrow a_1 = a_2$

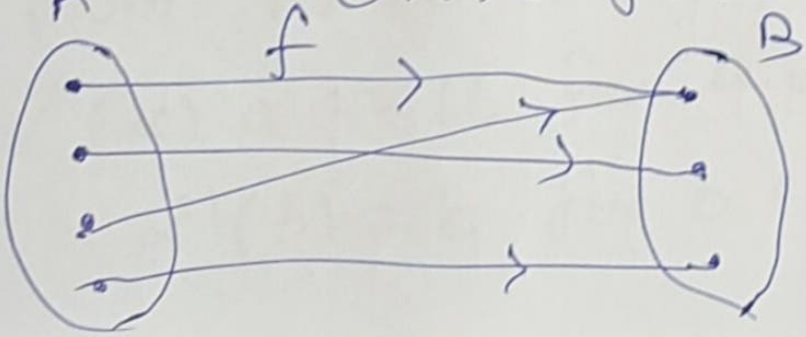
ψ is not one to one function

Definition from A Onto B.

A function ϕ from set A to a set B is said to be onto B. if each element of set B has at least one image of set A.

$\phi : A \longrightarrow B$ is onto if each element of B in B there is atleast one element a in A. $\Rightarrow \phi(a) = b$, $\boxed{x = a}$

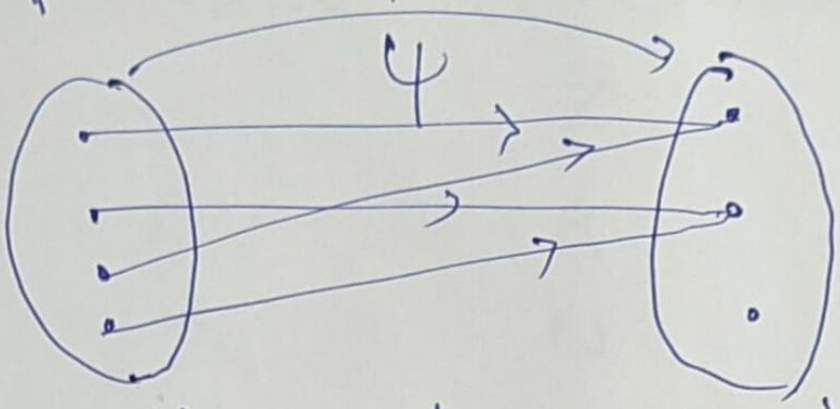
Onto function.



onto.

ϕ is an onto.

ψ is not onto function.



ψ is not an onto function.

PROPERTIES of FUNCTIONS

Given function

$\alpha : A \longrightarrow B$, $\beta : B \longrightarrow C$ and $\gamma : C \longrightarrow D$ then.

(1.) $\gamma(\beta\alpha) = (\gamma\beta)\alpha$ (Associative.)

(2) If α and β are one to one, then $\beta\alpha$ is also one to one

(3.) If α and β are onto then. $\beta\alpha$ is also onto.

(4) If α is one-to-one and onto then there is a function α^{-1} from B to A.

$\alpha^{-1}(\alpha)\, a = a$ for $a \in A$.

$\alpha\, \alpha^{-1}(b) = b$ for $b \in B$.

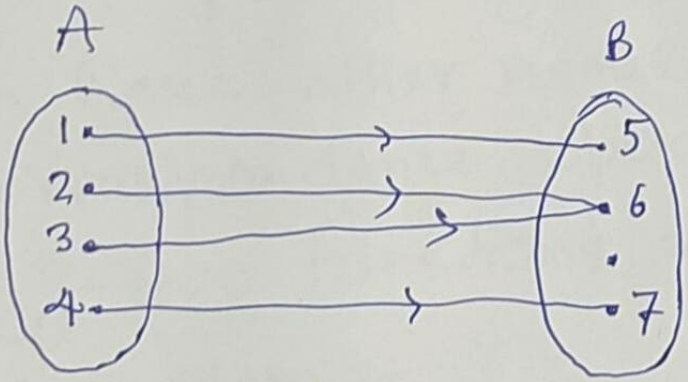
x DOMAIN	$y=f(x)$ CODOMAIN	Rule	1:1	onto
$\mathbb{Z}$	$\mathbb{Z}$	$x \longrightarrow x^3$	yes	no
$\mathbb{R}$	$\mathbb{R}$	$x \longrightarrow x^3$	yes	yes
$\mathbb{Z}$	$\mathbb{N}$	$x \longrightarrow \lvert x \rvert$	no	yes
$\mathbb{Z}$	$\mathbb{Z}$	$x \longrightarrow x^2$	no	no.

FUNCTIONS

FUNCTIONS: A function is a special case of relation. Let A and B be two non-empty sets and R be a relation from A to B, then R may not relate an element of set A to an element of set B. OR it may relate an element of set A to more then one element of B. But in function relate an element of A to a unique element of B.

Definition: Let A and B ~~sets~~ non-empty sets. A function f from A to B is a set of ordered pairs with the property that for each element x of A has a unique element y in B.

$$f: A \longrightarrow B \quad \text{or} \quad A \xrightarrow{f} B$$



Example 1: Let $A = \{1,2,3,4,5\}$, $B = \{0,1,2,3,5,7,9,12\}$

(i) $f = \{(1,1), (2,0), (3,7), (4,9), (5,12)\}$ f is a function from A to B because each element of A has a unique element in B. No element of A has two or more image in B.

Note:- Some element of B are not Associated with A.

(ii) $f = \{(1,3), (2,3), (3,5), (4,9), (5,9)\}$, then f is a function because each element of set A has a unique element of set B.

Range of f is $\{3, 5, 9\}$

(iii) $f = \{(1,1), (2,3), (4,7), (5,12)\}$ then f is not a function from A to B because the element 3 of set A has not image in Set B

(iv) $f = \{(1,1), (2,3), (3,5), (3,7), (4,9)\}$ then f is not a function because the different pairs (3,5), (3,7) have same First component.

If $f(x) = x^2$ then $f - R \to R$, this is Called numeric function -

function represent by f, g, h, Ψ, Φ - function have other names transform, map, correspondance. operations.

f and g two functions are equal

① dom f = dom g

② Range f = Range of g

③ $f(x) = g(x)$, $x \in$ dom f

$f(x) = g(a)$

Difference between relation and function.

1. A relation R from set A to set B is a subset of A×B, domain of R is the subset of A

But in a function from A to B domain of f must be equal to set A.

2. In a relation, one element of A may be related to more than one element of B

But in function one or more element of A must be related to only and only one element of B.

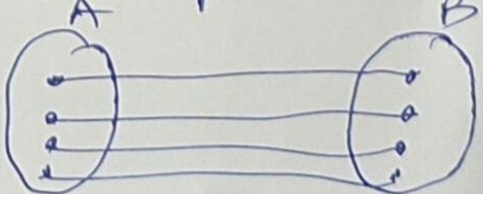
3. In a relation, some element of A may not be ~~any~~ related to any element of B.

So Every function is a relation.
But Every Relation is not a function

TYPES OF FUNCTIONS:

1. One-to One function: A function from A to B is one to one or injective distinct elements of A maped with distinct elements of B.

(injective)



Ex 2. Let $A = \{1, 2, 3\}$, $B = \{a, b, c, d\}$

Let $f(1) = a$, $f(2) = c$, $f(3) = d$

Here 1, 2, 3 elements of A are associated with different element of B.

Ex 3. $f(x) = 3x^2 - 1 \Rightarrow f(x_1) = 3x_1^2 - 1$

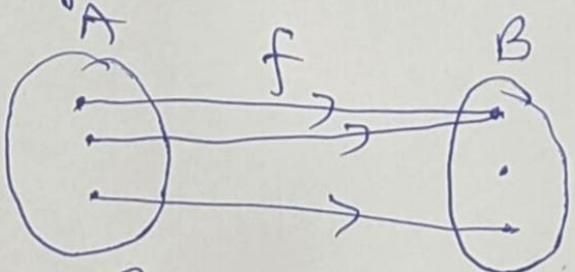
$f(x_2) = 3x_2 - 1$

$f(x_1) = f(x_2) \Rightarrow 3x_1 - 1 = 3x_2 - 1 \Rightarrow \boxed{x_1 = x_2}$

$f: A \longrightarrow B$, no of elements of set A may less or equal to no of element of B.

Second TYPE MANY ONE function

A function f from A to B is said to many-one function if two or more element of A has same image in B.



$f(x) = x^2$, $x = 1$, $f(1) = 1$

$x = -1$, $f(-1) = 1$

$x = 2$, $f(2) = 4$

$x = -2$, $f(-2) = 4$

